

From Sufficient Condition to Mean Square Consistency

If $E(\hat{\theta}_n) \rightarrow \theta$ and $Var(\hat{\theta}_n) \rightarrow 0$
then

$$E[(\hat{\theta}_n - \theta)^2] = Var(\hat{\theta}_n) + (E(\hat{\theta}_n) - \theta)^2 \rightarrow 0$$

We call that **L2 (Mean Square) Consistency**:

$$\hat{\theta}_n \xrightarrow{L^2} \theta$$

Strength (Strongest to Weakest):

$$\hat{\theta}_n \xrightarrow{a.s.} \theta \quad \text{or} \quad \hat{\theta}_n \xrightarrow{L^2} \theta \quad \Rightarrow \quad \hat{\theta}_n \xrightarrow{P} \theta \quad \Rightarrow \quad \hat{\theta}_n \xrightarrow{d} \theta$$

Consistency of the Sample Mean

Suppose

$$X_1, \dots, X_n \stackrel{iid}{\sim} (\mu, \sigma^2)$$

and consider the sample mean

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

We have

$$E(\bar{X}) = \mu, \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n} \rightarrow 0$$

Then, by Chebyshev's inequality,

$$P(|\bar{X} - \mu| > \varepsilon) \leq \frac{\sigma^2/n}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2} \rightarrow 0, \quad \forall \varepsilon > 0$$

Therefore

$$\bar{X} \xrightarrow{P} \mu$$

Consistency of the Sample Variance

Consider the sample variance

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

We have

$$E(S^2) = \sigma^2, \quad \text{Var}(S^2) \rightarrow 0$$

Then, by Chebyshev's inequality,

$$P(|S^2 - \sigma^2| > \varepsilon) \leq \frac{E[(S^2 - \sigma^2)^2]}{\varepsilon^2} = \frac{MSE(S^2)}{\varepsilon^2}$$

Since $E(S^2) = \sigma^2$, S^2 is unbiased, so

$$P(|S^2 - \sigma^2| > \varepsilon) \leq \frac{\text{Var}(S^2)}{\varepsilon^2} \rightarrow 0, \quad \forall \varepsilon > 0$$

Therefore

$$S^2 \xrightarrow{P} \sigma^2$$